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EXP8:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main() {
    char key[50], msg[200];
    char cipher[27], used[26] = {0};
    int i, k = 0;

    printf("Enter keyword: ");
    scanf("%s", key);
    printf("Enter plaintext: ");
    scanf("%s", msg);

    /* keyword first (without repeats), then unused letters in normal order */
    for (i = 0; key[i]; i++) {
        int c = toupper(key[i]) - 'A';
        if (!used[c]) { used[c] = 1; cipher[k++] = c + 'A'; }
    }
    for (i = 0; i < 26; i++)
        if (!used[i]) cipher[k++] = i + 'A';
    cipher[k] = '\0';

    printf("\nplain : a b c d e f g h i j k l m n o p q r s t u v w x y z\n");
    printf("cipher: ");
    for (i = 0; i < 26; i++) printf("%c ", cipher[i]);
    printf("\n\n");
```

```
    printf("Ciphertext: ");
    for (i = 0; msg[i]; i++) putchar(cipher[tolower(msg[i]) - 'a']);
    printf("\nDecrypted : ");
    for (i = 0; msg[i]; i++) {
        int j, ch = toupper(msg[i]);
        char c2 = cipher[tolower(msg[i]) - 'a'];
        for (j = 0; j < 26; j++) if (cipher[j] == c2) putchar(j + 'a');
    }
    printf("\n");
    return 0;
}
```

OUTPUT:

```
Enter keyword: CIPHER
Enter plaintext: meetmeatnoon

plain : a b c d e f g h i j k l m n o p q r s t u v w x y z
cipher: C I P H E R A B D F G J K L M N O Q S T U V W X Y Z

Ciphertext: KEETKECTLMML
Decrypted : meetmeatnoon
```